

# Thursday Warm-up: Electric Fields in Matter

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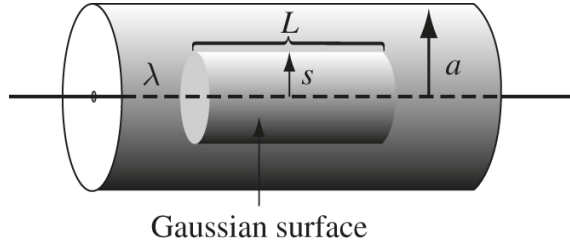


Figure 1: A charge distribution with density  $\lambda$  [C m<sup>-1</sup>] is embedded in a dielectric material with  $\mathbf{P} = (k/s)\hat{\mathbf{s}}$ .

## 3 Polarization and Bound Charge

1. Consider Fig. 1. Given  $\mathbf{P} = (k/s)\hat{\mathbf{s}}$ , calculate (a) the bound volume charge density,  $\rho_b$ , and (b) the bound surface charge density,  $\sigma_b$ .

## 1 Memory Bank

- **Bound surface charge density**, given a polarization  $\mathbf{P}$ :

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} \quad (1)$$

- **Bound volume charge density**, given a polarization  $\mathbf{P}$ :

$$\rho_b = -\nabla \cdot \mathbf{P} \quad (2)$$

- **Cartesian and cylindrical unit vectors:**

$$\hat{\mathbf{x}} = \cos \phi \hat{\mathbf{s}} - \sin \phi \hat{\phi} \quad (3)$$

$$\hat{\mathbf{y}} = \sin \phi \hat{\mathbf{s}} + \cos \phi \hat{\phi} \quad (4)$$

$$\hat{\mathbf{z}} = \hat{\mathbf{z}} \quad (5)$$

- **Divergence in cylindrical coordinates:**

$$\nabla \cdot \mathbf{v} = \frac{1}{s} \frac{\partial}{\partial s}(sv_s) + \frac{1}{s} \frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z} \quad (6)$$

- **The electric displacement:**

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad (7)$$

- **Gauss' Law for electric displacement:**

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,enc} \quad (8)$$

## 2 Gauss' Law Review

1. Consider Fig. 1. Imagine that  $\mathbf{P} = 0$  in the region surrounding the linear charge distribution. (a) Calculate  $\mathbf{E}$ . (b) Calculate  $V(\mathbf{r})$ .

## 4 The Electric Displacement

1. (a) Write down the differential form of Gauss' Law, and substitute  $\rho = \rho_f + \rho_b$ . (b) Substitute  $\mathbf{P}$  for  $\rho_b$  using the definition in the Memory Bank. (c) Solve for  $\rho_f$ . (d) Identify  $\mathbf{D}$ . What are the units of  $\mathbf{D}$ ?
2. (a) Calculate the *electric displacement* for the system in Fig. 1, inside the dielectric cylinder. First, use Gauss' Law for  $\mathbf{D}$ . Next, find  $\mathbf{E}$  using  $\mathbf{P}$  and  $\mathbf{D}$ . (b) Using the result for  $\sigma_b$  obtained in Sec. 3, find the field outside the cylinder using Gauss' Law. (c) Find  $\mathbf{D}$  outside the cylinder using Gauss' law, and check that it is consistent with (b).
3. Suppose the polarization with a volume below the xy-plane is  $\mathbf{P} = P\hat{\mathbf{z}}$ . (a) Find  $\sigma_b$  and  $\rho_b$ . (b) Find  $\mathbf{D}$  and  $\mathbf{E}$  in the region above the plane. (c) How would you conceive of  $\mathbf{E}$  and  $\mathbf{D}$  for a capacitor fill with a dielectric material?